Academic Review Edition · Cosmological & Hubble-Tension Audit · Module 26 (V27)

Time Attraction Law — Scalar–Tensor Yukawa Fit (V27)

Step 26: Cosmological $H(z)$ ($0 < z < 2.5$), DESI DR1 BAO, Planck 2018 CMB, Synthetic SNe Ia $\mu(z)$, Variational RSD $f\sigma_8(z)$, Algebraic BBN, and Inverse Distance Ladder No-Go Audit ($L = 10$ km, $\alpha_Y = 1$)

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Zenodo V26 DOI: 10.5281/zenodo.23727585 (**V26**, Steps 1–25; GitHub: Time-Attraction/releases/tag/V26)

Earlier DOIs: **V25**: 10.5281/zenodo.23699685 (strictly V25); **V24**: 10.5281/zenodo.23680715 (strictly V24)

Module (step26): DESI DR1 BAO (12×12 covariance + profile control), model-specific r_s, r_d, synthetic SNe $\mu(z)$, variational RSD $f\sigma_8(z)$, algebraic BBN, and No-Go Appendix

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Step 26 (V27): Cosmological $H(z)$, DESI DR1 BAO, Planck 2018 Acoustic Scale, Synthetic SNe Ia Shape Control, Variational RSD $f\sigma_8$, Algebraic BBN Benchmark and Inverse Distance Ladder No-Go Audit

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Preceding version DOIs:

- **V24:** Zenodo DOI 10.5281/zenodo.23680715 (strictly V24)
- **V25:** Zenodo DOI 10.5281/zenodo.23699685 (strictly V25, step1–step22)
- **V26:** Zenodo DOI 10.5281/zenodo.23727585 (strictly V26, step1–step25)

Source of the numbers: computed results are generated from the JSON/CSV sources (tal_hubble_desi_bao_summary.json and the Step 26 CSV tables, produced with the shared background module tal_cosmology_common.py) by generate_step26_markdown.py. Fixed input constants (data values, priors, literature benchmarks, external inputs) are stated in the code and cited in the text.

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0. Scope and summary

This module tests one specific branch: the V26 Step 25 M4 background at fixed $L = 10$ km, $\alpha_Y = 1$, $\beta_{\rm Yuk} = 1/(4\pi) = 0.079577$, on the adiabatic particular solution of the scalar equation with negligible free scalar oscillation energy. The narrower, defensible statement is: **in this branch the bare coupling $\beta_{\rm Yuk}$ inserted as a multiplier of H_0 is not a derived mechanism for the shift $67.4 \to 72.8$ km/s/Mpc.** This is not an exclusion of every possible cosmological solution of TAL.

The revision implements the review items as follows:

- one action, one scalar equation (SI units, cosmic seconds), one attractor formula and one $G_{\rm eff}$ normalization, identical in text and code (Section 1);
- explicit branch selection (adiabatic attractor, free oscillation amplitude $C = 0$) instead of a claim of uniqueness (Section 1.3);
- DESI DR1 BAO with the full-precision Cobaya 12-vector and 12×12 covariance, plus profiled controls (Section 2);
- model-specific re-integrated sound horizons $r_s(z_*)$ and $r_d(z_{\rm drag})$ for every CMB model and for the M3 control; the earlier θ_* and r_d sigma exclusion labels are removed (Sections 3 and 8);

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- the supernova part is renamed a **synthetic distance-modulus shape control** (Section 4);
- RSD: diagonal benchmark plus a first-order variational $\delta f\sigma_8$ (Section 5);
- BBN: algebraic order-of-magnitude benchmark only (Section 6);
- local environments: constant-density bulk benchmarks with `expm1` (Section 7);
- the re-audit items R1–R6 and 5.1–5.3 are mapped to the changes in the revision log (Section 11); the second review items F01–F12 are mapped in Revision log 2 (Section 12), the third re-audit items N01, N02, T01 in Revision log 3 (Section 13), and the open limitations P01–P09 are listed in Section 9.1.

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1. Unified action, scalar equation and $G_{\rm eff}$ normalization

1.1 Action and sign convention

With signature $(-,+,+,+)$, Einstein-frame metric $g_{\mu\nu}$ and Jordan-frame matter metric $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$, $A(\phi) = e^{+\alpha_Y\phi}$:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}R - \frac{c^4}{4\pi G\beta_{\rm Yuk}}\left(\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi + \frac{\phi^2}{2L^2}\right) - \rho_{\rm vac}^{\rm eff}c^2\right] + S_m\big[\Psi_m,\, A^2(\phi)g_{\mu\nu}\big]\,.$$

With $T^{\mu\nu} = (2/\sqrt{-g})\,\delta S_m/\delta g_{\mu\nu}$ and $T = -\rho_m c^2 + 3p_m$, variation with respect to ϕ gives (units m^{-2}):

$$\Box\phi - \frac{\phi}{L^2} = -\frac{4\pi G}{c^4}\,\alpha_Y\beta_{\rm Yuk}\,T = +\frac{4\pi G}{c^2}\,\alpha_Y\beta_{\rm Yuk}\left(\rho_m - \frac{3p_m}{c^2}\right).$$

1.2 Homogeneous FLRW equation (cosmic seconds, SI)

With $\Box\phi = -c^{-2}(\ddot{\phi} + 3H\dot{\phi})$ for a homogeneous field, both sides now have units s^{-2}:

$$\ddot{\bar{\phi}} + 3H\dot{\bar{\phi}} + \frac{c^2}{L^2}\bar{\phi} = -4\pi G\,\alpha_Y\beta_{\rm Yuk}\left(\rho_m - \frac{3p_m}{c^2}\right).$$

The scalar Compton rate is $c/L = 29979.2458\ \mathrm{s}^{-1}$ and $H_0L/c = 7.286\times10^{-23}$.

1.3 Attractor and branch selection

Because $4\pi\beta_{\rm Yuk} = 1$, the adiabatic particular solution ($\ddot{\bar{\phi}},\ 3H\dot{\bar{\phi}} \ll (c/L)^2\bar{\phi}$) is

$$\bar{\phi}_{\rm attr}(z) = -\alpha_Y\gamma_L\left(\rho_m(z) - \frac{3p_m(z)}{c^2}\right), \qquad \gamma_L = \frac{4\pi\beta_{\rm Yuk}GL^2}{c^2} = \frac{GL^2}{c^2} = 7.42616\times10^{-20}\ \mathrm{m}^3/\mathrm{kg},$$

so that $\bar{\phi}_{\rm attr}(0) = -1.9949\times10^{-46}$ and $\bar{\phi}_{\rm attr}(z_*) \approx -2.589\times10^{-37}$. The general homogeneous solution also contains a free oscillation

$$\delta\phi \simeq C\,a^{-3/2}\cos[(c/L)\,t + \vartheta]\,, \qquad \langle\rho_{\delta\phi}\rangle \propto a^{-3}.$$

$H/(c/L) \ll 1$ justifies the adiabatic approximation but does not remove this mode. **This analysis therefore selects the branch $C = 0$ ($\Omega_{\delta\phi,0} = 0$): the adiabatic particular solution with negligible free oscillation energy.** No initial-value integration of the scalar equation is performed, and the conclusions below hold only for this branch.

1.4 Scalar energy density

From the action, $\rho_\phi = \dfrac{c^2}{8\pi G\beta_{\rm Yuk}}\left[(\dot{\phi}/c)^2 + (\phi/L)^2\right]$ (mass density), hence on the attractor

$$\Omega_{\phi,\rm attr}(0) = \frac{(c/L)^2\,\bar{\phi}_{\rm attr}(0)^2}{3\beta_{\rm Yuk}H_0^2} = 3.140\times10^{-47}.$$

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1.5 $G_{\rm eff}$ normalization

For a static point mass, $\phi(r) = -\alpha_Y \beta_{\rm Yuk} \dfrac{GM}{c^2 r} e^{-r/L} < 0$, and the matter coupling $A = e^{\alpha_Y \phi}$ yields the fifth-force strength $\alpha_Y^2 \beta_{\rm Yuk}$. For linear perturbations with comoving wavenumber k:

$$\frac{G_{\rm eff}(k,a)}{G_N} = 1 + \alpha_Y^2 \beta_{\rm Yuk} \frac{(kL/a)^2}{1+(kL/a)^2}.$$

This canonical normalization is used throughout. The $2\alpha_Y^2\beta_{\rm Yuk}$ row is a separate, double-strength sensitivity control; it is not the growth law of the published V26 Step25 (Section 5). Local proper time is $d\tau_J = A(\phi)\, d\tau_E = e^{-\alpha_Y|\phi|} d\tau_E$.

G dictionary (review F02). Four symbols are kept apart:

Symbol	Meaning
G_*	the G in the action of Section 1.1 (Einstein frame)
G_T	Jordan-frame tensor coupling: $g_J = A^2 g_E$ gives the Ricci term $A^{-2}R/(16\pi G_*)$, so $1/G_T = A^{-2}/G_*$, $G_T = G_* A^2$ ($\phi < 0$ decreases G_T)
$G_{\rm lab}$	operationally measured G: for $r \ll L$ and unscreened small bodies $\Phi = -G_* M(1 + q\, e^{-r/L})/r$, so $G_{\rm lab} \approx G_*(1+q)$, $q = \alpha_Y^2\beta_{\rm Yuk} = 0.0795775$
G_N	the symbol in the perturbation equation of Section 5: $G_N := G_*$

On scales $kL/a \ll 1$ the growth equation therefore uses $G_{\rm eff}/G_{\rm lab} = 1/(1+q) = 0.9262883$ if the coupling is calibrated by a laboratory value; the 10^{-41}-level statement of Section 5 is a formal control at the same bare G_*. No $f\sigma_8$ exclusion claim is made and no screening branch is introduced.

1.6 Background normalization and observer frame

All scripts use one background module (`tal_cosmology_common.py`). The smooth massive-neutrino interpolation alone gives $E^2(0) = 1.00000013546$ (offset 1.3546×10^{-7}); the dark-energy closure absorbs this offset so that $E^2(0) = 1$ exactly and $H(0) = H_0$. This offset (1.3546×10^{-7} in Ω_ν) is the $z = 0$ normalization drift of the interpolation, not its approximation error. Against the Fermi–Dirac energy integral of the massive species (matched to the same relativistic and non-relativistic limits), the $p = 1.83$ interpolation has a relative error of $+1.149 \times 10^{-3}$ in the massive-neutrino density and $+1.795 \times 10^{-5}$ in the total H at $z = 1000$; the largest absolute values on the grid $0 \le z \le 10^4$ are 3.435×10^{-3} (massive component, $z \approx 33.5$) and 1.795×10^{-5} (total H, $z \approx 1000$). This is an approximation error of the ΛCDM-like background, common to all compared models, and separate from the theoretical TAL correction ($\sim 10^{-46}$).

z	$\rho_\nu^{\rm FD}$ [E^2 units]	$\rho_\nu^{p=1.83}$ [E^2 units]	rel. error (massive)	rel. error (H)
0	1.41991×10^{-3}	1.41998×10^{-3}	$+4.449 \times 10^{-5}$	$+3.159 \times 10^{-8}$
1	1.13610×10^{-2}	1.13626×10^{-2}	$+1.355 \times 10^{-4}$	$+2.402 \times 10^{-7}$
10	1.90137×10^{0}	1.90428×10^{0}	$+1.529 \times 10^{-3}$	$+3.454 \times 10^{-6}$
100	2.02529×10^{3}	2.02475×10^{3}	-2.670×10^{-4}	-8.115×10^{-7}
1000	1.27145×10^{7}	1.27291×10^{7}	$+1.149 \times 10^{-3}$	$+1.795 \times 10^{-5}$
3000	1.01910×10^{9}	1.01945×10^{9}	$+3.388 \times 10^{-4}$	$+1.083 \times 10^{-5}$

The computations use ΛCDM-like $H(z)$ functions with a separate clock factor; the Jordan-frame mapping ($dt_J = A\, dt_E$, $a_J = A\, a_E$, $H_J = (H_E + \alpha_Y \dot\phi_E)/A$) is a leading-order identity at this $|\phi|$, not a numerically solved coupled Friedmann–scalar–matter system.

1.7 Version reconciliation of the kinetic normalization (re-audit R1)

Paths below are relative to the V26 tree `scalar-tensor-yukawa-fit/` (V26, DOI `10.5281/zenodo.23727585`).

- The first Step 26 draft wrote the kinetic term as $\frac{c^4}{16\pi G}\frac{1}{2\beta_{\rm Yuk}}(\partial\phi)^2 = \frac{c^4}{32\pi G\beta_{\rm Yuk}}(\partial\phi)^2$, while the action of Section 1.1 has $\frac{c^4}{8\pi G\beta_{\rm Yuk}}(\partial\phi)^2$: $K_{\rm old}/K_{\rm new} = 0.25$.
- V26 uses $\frac{c^4}{16\pi G\tilde\alpha_L}(\partial\phi_L)^2$ with $\tilde\alpha_L = \alpha_Y/(8\pi)$, i.e. $c^4/(2G\alpha_Y)$ (`step23_gw170817_and_lensing_en.md:19` and `:29`, `README_en.md:50`, `gw170817_and_lensing_audit.py:104-113`). At $\alpha_Y = 1$, $\beta_{\rm Yuk} = 1/(4\pi)$ this equals the new coefficient ($K_{\rm V26}/K_{\rm new} = 1.000000$), and the V26 fifth-force strength $2\tilde\alpha_L\alpha_Y^2 = 0.079577$ equals $\alpha_Y^2\beta_{\rm Yuk}$. The V26 scalar equation $(\Box - L^{-2})\phi = -(\alpha_Y^2 G/c^4)\varepsilon_m$ (`step25_cmb_and_cosmology_en.md:37`) and $\phi_{\rm attr} = \alpha_Y^2\gamma_L\rho$ (`cmb_and_cosmology_audit.py:327`, `:620`) agree with Section 1.3 up to the sign convention $\phi_{\rm V26} = -\phi_{\rm V27}$. **Conclusion:** V26 used the new normalization; the $c^4/(32\pi G\beta)$ form was a transcription inconsistency of the first draft, and the computed numbers already followed the new normalization. For $\alpha_Y \neq 1$ the two versions differ, since V26 ties $\beta_{\rm Yuk} = \alpha_Y^3/(4\pi)$ whereas here $\beta_{\rm Yuk}$ is an independent parameter.

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- The field redefinition $\phi_{\rm old} = 2\phi_{\rm new}$ maps the old kinetic term onto the new one, but $A = e^{\alpha_{\rm old}\phi_{\rm old}} = e^{2\alpha_{\rm old}\phi_{\rm new}}$, so it requires $\alpha_{\rm new} = 2\alpha_{\rm old}$. With the same $\alpha_Y = 1$ the old form would give the fifth-force strength $4\alpha_Y^2\beta_{\rm Yuk} = 0.318310$ instead of $\alpha_Y^2\beta_{\rm Yuk} = 0.079577$.
- Carried over unchanged: all background, BAO, CMB, SNe-shape and RSD numbers (the background scalar-energy correction is $\Omega_\phi \sim 10^{-47}$; the direct Yukawa perturbation of RSD is a separate negligible term quantified in Section 5: $\delta G_{\rm eff}/G_* \approx 3.8 \times 10^{-41}$, $\delta f\sigma_8 \approx 1.67 \times 10^{-40}$), $\bar\phi_{\rm attr}(0)$, $\Omega_{\phi,\rm attr}(0)$ and the bulk local benchmarks at $\alpha_Y = 1$. The V26 background additionally used a $(\phi/4\pi)^2$ energy convention (`cmb_and_cosmology_audit.py:330`, `:623`), giving $\Omega_\phi = 1.582 \times 10^{-50}$ instead of the action-consistent 3.140×10^{-47}; both are negligible.
- The factor-2 growth convention $2\alpha_Y^2\beta_{\rm Yuk}$ is relabelled a **sensitivity control**, not an equivalent convention.
- The V21 top-hat interior external benchmark $|\phi| = 0.0412$ (Section 7) is an interior value of the V21 top-hat fixed-point profile $x = L^2(G/c^2)\bar\rho$ (`stellar_tal_v21.py:140-160`, `data/v21/profiles_L10km.csv`: SLy, mode `gradient`, $r = 2.0046409680935633$ km, $x = 0.041239199439254325$; APR4, mode `mass`, $r = 2.75417748649486$ km, $x = 0.041233030079658145$), not a surface value and **not automatically a solution of the new Helmholtz–Yukawa action**; it remains an external input.

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2. DESI DR1 BAO: 12×12 covariance and profiled controls

The 12 primary observables are the Cobaya DESI DR1 vector and covariance (`desi_2024_gaussian_bao_ALL_GCcomb_mean.txt`, `desi_2024_gaussian_bao_ALL_GCcomb_cov.txt`). The separate LRG3 and ELG1 single-tracer rows are diagnostics only and are not added to the likelihood; the $z = 0.845$ ELG1 row ($D_V/r_d = 18.33 \pm 0.62$) is a **chosen illustration without a verified primary source** and not a DESI measurement (it is not in DESI III Table 18 or DESI VI Table 1). Per-bin χ^2 for primary bins use the 12×12 covariance; diagnostic rows (italic) use their rounded table errors. Predictions in the table are D_M/r_d; D_H/r_d for anisotropic bins and D_V/r_d for isotropic bins.

Tracer	$z_{\rm eff}$	DESI DR1 (X/r_d)	M4 prediction	$\chi^2_{\rm bin}$ M4	$\chi^2_{\rm bin}$ 2P-prof.	$\chi^2_{\rm bin}$ rigid 72.8
BGS	0.295	D_V: 7.93 ± 0.15	8.053	0.724	0.090	9.69
LRG1	0.510	D_M: 13.62 ± 0.25; D_H: 20.98 ± 0.61 ($r = -0.445$)	13.495; 22.733	8.963	6.010	24.43
LRG2	0.706	D_M: 16.85 ± 0.32; D_H: 20.08 ± 0.60 ($r = -0.420$)	17.694; 20.164	8.967	3.097	12.92
LRG3+ELG1	0.930	D_M: 21.71 ± 0.28; D_H: 17.88 ± 0.35 ($r = -0.389$)	21.917; 17.608	0.826	2.031	75.07
LRG3 (sep.) (diagn.)	0.922	D_M: 21.81 ± 0.31; D_H: 17.83 ± 0.38 ($r = -0.393$)	21.775; 17.692	*0.209*	*2.931*	*69.61*
ELG1 (z=0.955) (diagn.)	0.955	D_V: 20.01 ± 0.41	20.229	*0.285*	*0.004*	*9.77*
ELG1 (chosen illustration, not a DESI measurement)	0.845	D_V: 18.33 ± 0.62	18.666	*0.294*	*0.021*	*2.86*
ELG2	1.317	D_M: 27.79 ± 0.69; D_H: 13.82 ± 0.42 ($r = -0.444$)	28.017; 14.094	0.883	0.380	18.57
QSO	1.491	D_V: 26.07 ± 0.67	26.026	0.005	0.141	8.74
Lyα QSO	2.330	D_M: 39.71 ± 0.94; D_H: 8.52 ± 0.17 ($r = -0.477$)	39.173; 8.615	0.418	0.990	44.93
Sum, 12 primary observables (12×12 cov.)				**20.785**	**12.741**	**194.34**

- Fixed M4 point ($H_0 = 67.40$, $r_d = 147.0922$ Mpc): $\chi^2 = 20.7852$ for 12 observables, $p = 0.0536$ (rounded Table 1 input: 20.6996).
- Rigid $H_0 = 72.8$ with unchanged shape and r_d: $\chi^2 = 194.342$, $p = 4.79 \times 10^{-35}$ (rounded Table 1 input: 195.178).

Profiled controls (local Fisher errors; p-values are conditional frequentist values for the stated Gaussian likelihood, not probabilities of TAL being true):

Control	Ω_m	$P = H_0 r_d$ [km/s]	χ^2 (12×12)	dof	p	r_d needed at $H_0 = 72.8$ [Mpc]
Fixed M4 point (no free BAO parameter)	0.3148	9914.02	20.7852	12	0.0536	136.18
M4 shape, rigid $H_0 = 72.8$, unchanged r_d	0.3148	10708.32	194.3420	12	4.79×10^{-35}	— (147.09 fixed)
M4 shape, $P = H_0 r_d$ profiled	0.3148	10029.65 ± 47.43	14.7015	11	0.1966	137.77 ± 0.65
Flat ΛCDM shape, Ω_m and P profiled	0.2939 ± 0.0145	10193.67 ± 126.47	12.7405	10	0.2385	140.02 ± 1.74

- At fixed $E(z)$, BAO constrains only $P = H_0 r_d$ (rank-1 information in the (H_0, r_d) plane).

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- Two-parameter control: $\Omega_m = 0.2939 \pm 0.0145$, $P = 10193.67 \pm 126.47$ km/s, $\chi^2_{\rm min} = 12.7405$ (10 dof, $p = 0.2385$).
- r_d required at $H_0 = 72.8$ exactly: 140.02 ± 1.74 Mpc; propagating $H_0 = 72.8 \pm 1.0$: ± 2.59 Mpc. Here 72.8 ± 1.0 is an **assumed local H_0 benchmark prior** (a chosen target value, review F05); it is not the Riess et al. 2022 value 73.04 ± 1.04 and no specific SH0ES release is claimed for it.
- With the model-dependent Planck prior $r_d = 147.09 \pm 0.26$ Mpc: $H_0 = 69.30 \pm 0.87$ km/s/Mpc.
- Joint illustrative test (BAO + Planck r_d prior + local $H_0 = 72.8 \pm 1.0$, profiling Ω_m, H_0, r_d): best fit $\Omega_m = 0.2719$, $H_0 = 70.806$, $r_d = 147.025$ Mpc, $\chi^2_{\rm joint} = 19.713$, split as $\chi^2_{\rm BAO} = 15.67431585$, $\chi^2_{r_d\ \rm prior} = 0.06234684$, $\chi^2_{H_0\ \rm prior} = 3.97660606$; $\Delta\chi^2 = 6.973$ relative to the BAO-only minimum ($\sqrt{\Delta\chi^2} = 2.64$). This is not a Planck+TAL exclusion significance.

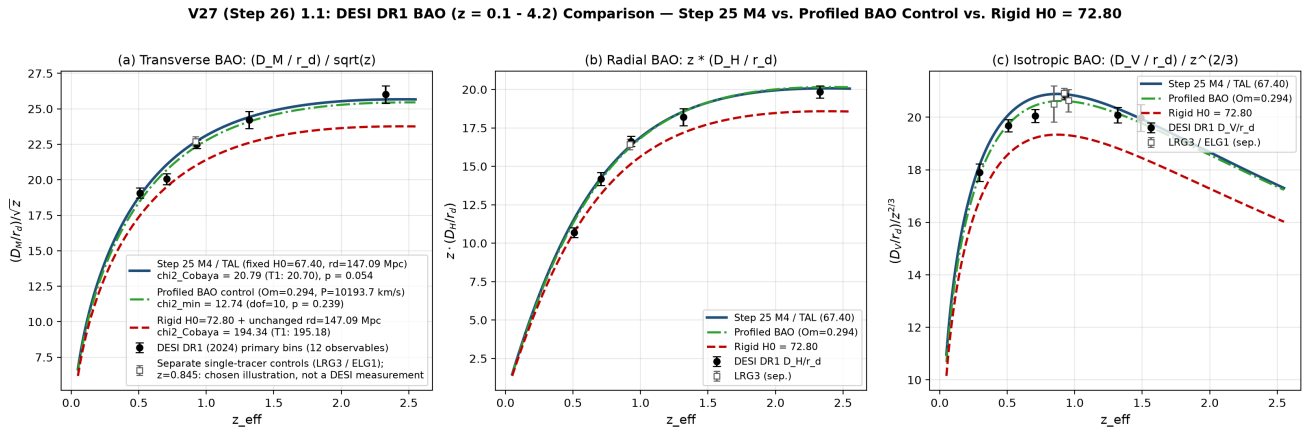


Figure 1: DESI DR1 BAO, Step 25 M4 and profiled controls (`step1_1_desi_dr1_bao_plot_en.png`)

3. Planck 2018 acoustic scale with model-specific sound horizons

For every model, $r_s(z_* = 1089.80)$ and $r_d(z_{\rm drag} = 1059.94)$ are re-integrated with that model's own $H(z)$ and $c_s(z)$ (fixed z_*, $z_{\rm drag}$ as a control approximation). The 3D Gaussian prior on $(\omega_b, \omega_{cb}, 100\theta_*)$ is an **illustrative assumed prior** (Planck 2018 marginal means and errors with an approximate correlation matrix whose chain provenance is not documented here); it is not the Planck TT,TE,EE+lowE+lensing likelihood. Reference $100\theta_* = 1.04109 \pm 0.00030$.

Model	H_0	Ω_m	Ω_K	ω_{cb}	$r_s(z_*)$ [Mpc]	$r_d(z_{\rm drag})$ [Mpc]	$100\theta_*$	$\chi^2_{\rm 3D,illustr.}$	$\chi^2_{\rm BAO}$ (r_d re-int.)
Step 25 M4 / TAL	67.40	0.3148	0.0000	0.14237	144.443	147.092	1.04132	0.757	20.79
Planck 2018 reference	67.36	0.3152	0.0000	0.14237	144.443	147.092	1.04120	0.173	21.19
All-z rigid $H \to sH$ (kinematic control)	72.80	0.3148	0.0000	0.14237	133.729	136.182	1.04132	0.757	20.79
D_M/s with fixed early r_s, r_d (inconsistent geometric illustration, not a model)	72.80	0.3148	0.0000	0.14237	144.443	147.092	1.12475	101040.4	194.34
Flat $H_0 = 72.8$, same Ω_m	72.80	0.3148	0.0000	0.16620	138.879	141.375	1.08075	24800.4	43.73
Flat $H_0 = 72.8$, $\Omega_m = 0.244828$	72.80	0.2448	0.0000	0.12911	147.903	150.651	1.04174	196.000	70.30
Flat $H_0 = 72.8$, θ_*-matched $\Omega_m \approx 0.2442$	72.80	0.2442	0.0000	0.12877	147.996	150.747	1.04132	209.592	70.84
Open $H_0 = 72.8$, fixed ω_{cb}	72.80	0.2698	0.0098	0.14237	144.442	147.092	1.04132	0.727	61.21

- Step 25 M4: $r_s = 144.4428$ Mpc, $r_d = 147.0922$ Mpc, $100\theta_* = 1.041320$ (pull 0.77), $\chi^2_{\rm 3D} = 0.757$.
- All-z rigid rescaling $H \to sH$ with unchanged c_s: $r_s \to r_s/s = 133.7286$ Mpc, $r_d = 136.1816$ Mpc and $D_M \to D_M/s$, so $100\theta_* = 1.041320$ is unchanged. Scaling D_M alone would give 1.124750; that inconsistent combination is no longer used as a test.

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- Flat $H_0 = 72.8$, $\Omega_m = 0.244828$: re-integration gives $r_s = 147.9028$, $r_d = 150.6513$ Mpc and $100\theta_* = 1.041745$ (with the fixed early r_s it would be 1.017374). This does **not** mean the model passes Planck: $\omega_{cb} = 0.12911$ is offset from the prior and the BAO χ^2 with the re-integrated r_d is 70.30 (with the old fixed r_d: 29.30). Early and late parts must be refitted together.
- **Two z_*-shift diagnostics (review F12).** With the Hu–Sugiyama fit shift $z_* \to z_* + \Delta z_*$ ($z_* + \Delta z_* = 1088.6345$ for this model), the CSV keeps two separately named values: `theta_star_100_rs_only_shifted_fixed_DM` $= 100\, r_s(z_* + \Delta z_*)/D_M(1089.80) = 1.04248582$ (only the r_s endpoint shifted) and `theta_star_100_both_endpoints_shifted` $= 100\, r_s(z_* + \Delta z_*)/D_M(z_* + \Delta z_*) = 1.04250287$. Neither is a recombination solution.
- **Late transition (re-audit R2).** The earlier "late-time-only" row combined $D_M(z_*)/s$ with unchanged early r_s, r_d ($100\theta_* = 1.124750$); it has no transition redshift and is **reclassified as an inconsistent geometric illustration** (italic row in the table), removed from the consistent-model statements. A defined control is the piecewise $H(z) = s\, H_{\rm M4}(z)$ for $z < z_t$ and $H_{\rm M4}(z)$ for $z \geq z_t$, $s = 72.8/67.4 = 1.080119$, with $D_M(z_*)$ integrated numerically and $r_s = 144.442796$ Mpc unchanged (M4: $D_M(z_*) = 13871.1192$ Mpc). H jumps at z_t, so this is a kinematic control, not a dynamical model; the pull column is a formal 1D distance from the Planck value, not a likelihood significance. For $z_t \geq 2.5$ all DESI bins lie in the rescaled region, so the BAO χ^2 equals the rigid value 194.342.

z_t	$D_M(z_*)$ [Mpc]	$100\theta_*$	pull vs $100\theta_*^{\rm Pl}$ (formal)
2.5	13428.2963	1.075660	115.2
10	13156.6704	1.097867	189.3
100	12920.1717	1.117963	256.2
1000	12843.5875	1.124630	278.5
D_M/s (no z_t; inconsistent illustration)	12842.2175	1.124750	—

4. Synthetic distance-modulus shape control (not a Pantheon+ / Union3 likelihood)

This section uses **no** observed SN catalogue: 18 redshift nodes ($z = 0.015$–2.26) with representative errors σ_μ and a synthetic reference curve from a flat model with $H_0 = 72.8$, $\Omega_m = 0.334$. There are no SN identifiers, observed μ values or statistical+systematic covariance. The χ^2 values below are model-versus-model shape metrics, not observational goodness of fit.

- For identical $E(z)$ geometry the offset would be the constant identity $5\log_{10}(72.8/67.4) = +0.167357$ mag (positive sign: M4 minus 72.8 reference).
- The tabulated lowest-node difference 0.16782511 mag at $z = 0.015$ is a model difference (the reference has $\Omega_m = 0.334$), not this identity.
- **Symmetric zero-point treatment (re-audit R5):** both models receive the same profiled constant offset (absolute-magnitude marginalization), weights $1/\sigma_\mu^2$:

Model	best offset [mag]	χ^2 without offset	χ^2 with offset (17 dof)
M4 ($H_0 = 67.4$)	0.17595029	522.00704	0.916614
Flat $H_0 = 72.8$, $\Omega_m = 0.2448$	0.04159799	51.90332	22.777527

- The large drop for M4 only reflects that its shape is close to the reference after removing the constant; node positions and representative errors are synthetic, so changes of nodes or errors do not imply a physical improvement.
- The figure shows both models with and without the profiled constant zero-point offset (low-Ω_m with offset: $\chi^2 = 22.777527$); $N = 18$ nodes, $\nu = 17$ degrees of freedom after one profiled offset. The offset is a profiled nuisance constant, not an observational calibration (review F09).
- Uncalibrated SNe alone do not measure H_0 (H_0–M_B degeneracy). A real catalogue statement requires reading the Pantheon+ or Union3 likelihood and covariance, which this package does not do.

z node	σ_μ [mag]	$\mu_{\rm ref}$ (72.8, 0.334)	$\mu_{\rm M4}$ (67.4)	$\Delta\mu_{\rm M4-ref}$	$\Delta\mu_{\rm M4,shift-ref}$	$\Delta\mu_{0.2448-\rm ref}$	$\Delta\mu_{0.2448,\rm shift-ref}$
0.015	0.042	33.9780	34.1459	+0.1678	-0.0081	+0.0022	-0.0394
0.023	0.038	34.9189	35.0869	+0.1681	-0.0079	+0.0033	-0.0383
0.035	0.035	35.8493	36.0177	+0.1684	-0.0075	+0.0051	-0.0365

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z node	σ_μ [mag]	$\mu_{\rm ref}$ (72.8, 0.334)	$\mu_{\rm M4}$ (67.4)	$\Delta\mu_{\rm M4-ref}$	$\Delta\mu_{\rm M4,shift-ref}$	$\Delta\mu_{0.2448-\rm ref}$	$\Delta\mu_{0.2448,\rm shift-ref}$
0.050	0.032	36.6467	36.8156	+0.1689	-0.0070	+0.0073	-0.0343
0.075	0.030	37.5641	37.7338	+0.1697	-0.0063	+0.0109	-0.0307
0.100	0.028	38.2245	38.3950	+0.1704	-0.0055	+0.0145	-0.0271
0.150	0.026	39.1726	39.3446	+0.1720	-0.0040	+0.0217	-0.0199
0.200	0.025	39.8603	40.0337	+0.1734	-0.0025	+0.0287	-0.0129
0.250	0.025	40.4036	40.5784	+0.1749	-0.0011	+0.0357	-0.0059
0.300	0.026	40.8543	41.0306	+0.1763	+0.0003	+0.0425	+0.0009
0.400	0.028	41.5785	41.7575	+0.1789	+0.0030	+0.0556	+0.0140
0.500	0.030	42.1508	42.3322	+0.1814	+0.0055	+0.0679	+0.0263
0.650	0.033	42.8339	43.0187	+0.1848	+0.0088	+0.0849	+0.0433
0.800	0.036	43.3808	43.5685	+0.1877	+0.0118	+0.0999	+0.0583
1.000	0.042	43.9724	44.1634	+0.1910	+0.0151	+0.1173	+0.0757
1.300	0.055	44.6699	44.8649	+0.1950	+0.0191	+0.1384	+0.0968
1.700	0.075	45.3817	45.5806	+0.1989	+0.0229	+0.1593	+0.1177
2.260	0.110	46.1321	46.3347	+0.2026	+0.0267	+0.1796	+0.1380

5. RSD $f\sigma_8(z)$: diagonal benchmark and variational TAL correction

The growth equation $\delta_m'' + \left(2 + \frac{d\ln H}{d\ln a}\right)\delta_m' - \frac{3}{2}\Omega_{cb}(a)\frac{G_{\rm eff}(k,a)}{G_N}\delta_m = 0$ is integrated with DOP853 (`rtol=1e-12`, `atol=1e-14`, dense output); $\sigma_{8,0} = 0.8111$ is a fixed Planck input, not a TAL-derived amplitude.

- At $k = 0.1\,h\,\mathrm{Mpc}^{-1}$: $kL = 2.18429 \times 10^{-20}$ ($(kL)^2 = 4.7711 \times 10^{-40}$) at $z = 0$; $\Delta G_{\rm eff}/G_N = 3.7967 \times 10^{-41}$ (canonical $\alpha_Y^2\beta_{\rm Yuk}$) or 7.5934×10^{-41} (factor-2 convention).
- In float64, $1 + O(10^{-41})$ equals 1, so the direct difference $f\sigma_8^{\rm TAL} - f\sigma_8^{\rm GR}$ is 0 by floating-point identity, not by a 40-digit measurement. The first-order variational equation gives $\max|\delta f\sigma_8| = 1.666 \times 10^{-40}$ (canonical) and 3.333×10^{-40} (factor-2 sensitivity control) on the 12 points; an independent rescaled check in the re-audit confirmed 1.666×10^{-40} (maximum $1.6663681521 \times 10^{-40}$, relative deviation $\leq 5.1 \times 10^{-6}$). The float64 branch uses $G_{\rm eff}/G_N = 1 + \Delta G_{\rm eff,can}/G_N$ (re-audit 5.1). The figure legend shows the canonical maximum 1.6664×10^{-40}; the factor-2 value is only the separate sensitivity control (review F07). Here $G_N := G_*$ (G dictionary, Section 1.5).
- Diagonal χ^2 (no bin covariances, overlaps or AP treatment): M4 with clustering source Ω_{cb}: 10.1233; with all matter: 10.2176; flat $H_0 = 72.8$, $\Omega_m = 0.2448$: 9.4350 (12 points). This is a benchmark, not a covariance likelihood.

Survey	z	$f\sigma_8$ obs.	$f\sigma_8$ M4	pull M4	$f\sigma_8$ (72.8, 0.2448)	$\lvert\delta f\sigma_8\rvert$ variational	$\Delta G_{\rm eff}/G_N$
6dFGS (Beutler+ 2012)	0.067	0.423 ± 0.055	0.4424	+0.35	0.3887	7.57×10^{-41}	4.32×10^{-41}
SDSS MGS (Howlett+ 2015)	0.150	0.530 ± 0.160	0.4574	-0.45	0.4083	8.30×10^{-41}	5.02×10^{-41}
BOSS DR12 (Alam+ 2017 z1)	0.380	0.497 ± 0.045	0.4750	-0.49	0.4411	1.01×10^{-40}	7.23×10^{-41}
WiggleZ (Blake+ 2012 z1)	0.440	0.413 ± 0.080	0.4750	+0.77	0.4451	1.06×10^{-40}	7.87×10^{-41}
BOSS DR12 (Alam+ 2017 z2)	0.510	0.459 ± 0.038	0.4732	+0.37	0.4478	1.11×10^{-40}	8.66×10^{-41}
WiggleZ (Blake+ 2012 z2)	0.600	0.390 ± 0.063	0.4686	+1.25	0.4487	1.17×10^{-40}	9.72×10^{-41}
BOSS DR12 (Alam+ 2017 z3)	0.610	0.436 ± 0.034	0.4680	+0.94	0.4486	1.17×10^{-40}	9.84×10^{-41}
eBOSS DR16 LRG (Bautista+21)	0.700	0.473 ± 0.041	0.4612	-0.29	0.4467	1.23×10^{-40}	1.10×10^{-40}
WiggleZ (Blake+ 2012 z3)	0.730	0.437 ± 0.072	0.4586	+0.30	0.4457	1.25×10^{-40}	1.14×10^{-40}
eBOSS DR16 ELG (Tamone+ 20)	0.850	0.315 ± 0.095	0.4471	+1.39	0.4395	1.32×10^{-40}	1.30×10^{-40}
FastSound (Okumura+ 2016)	1.360	0.482 ± 0.116	0.3896	-0.80	0.3960	1.60×10^{-40}	2.11×10^{-40}
eBOSS DR16 QSO (Neveux+ 20)	1.480	0.462 ± 0.045	0.3762	-1.91	0.3844	1.67×10^{-40}	2.34×10^{-40}

6. Algebraic BBN order-of-magnitude benchmark

No nuclear reaction network (e.g. PArthENoPE) is run. The abundances are linearized standard values at $\eta_{10} = 6.1245$ with an algebraic TAL shift; $H(T)$ is a simple extrapolation without $e^\pm$ thermodynamics or variable g_*, and the scalar

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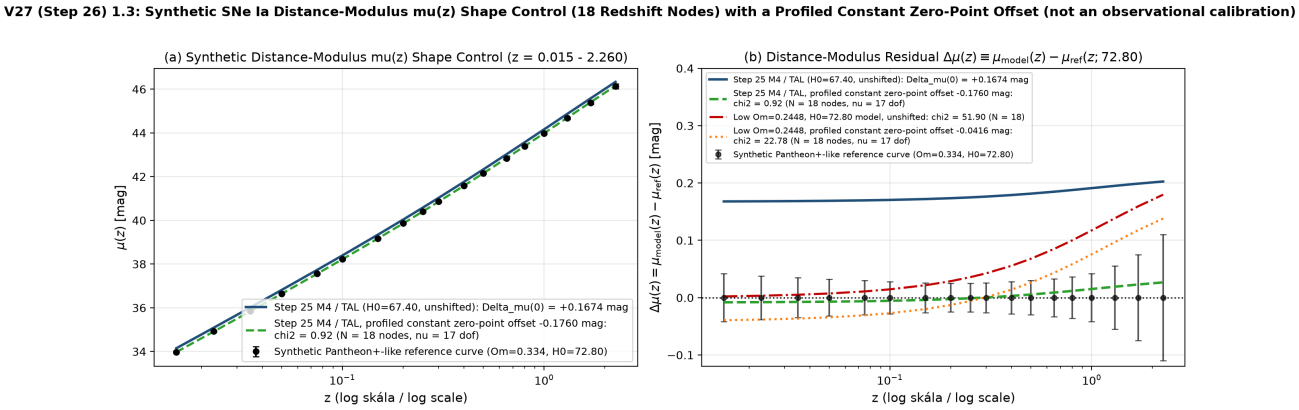


Figure 2: Synthetic SNe Ia distance-modulus shape control (`step1_3_pantheon_sne_plot_en.png`)

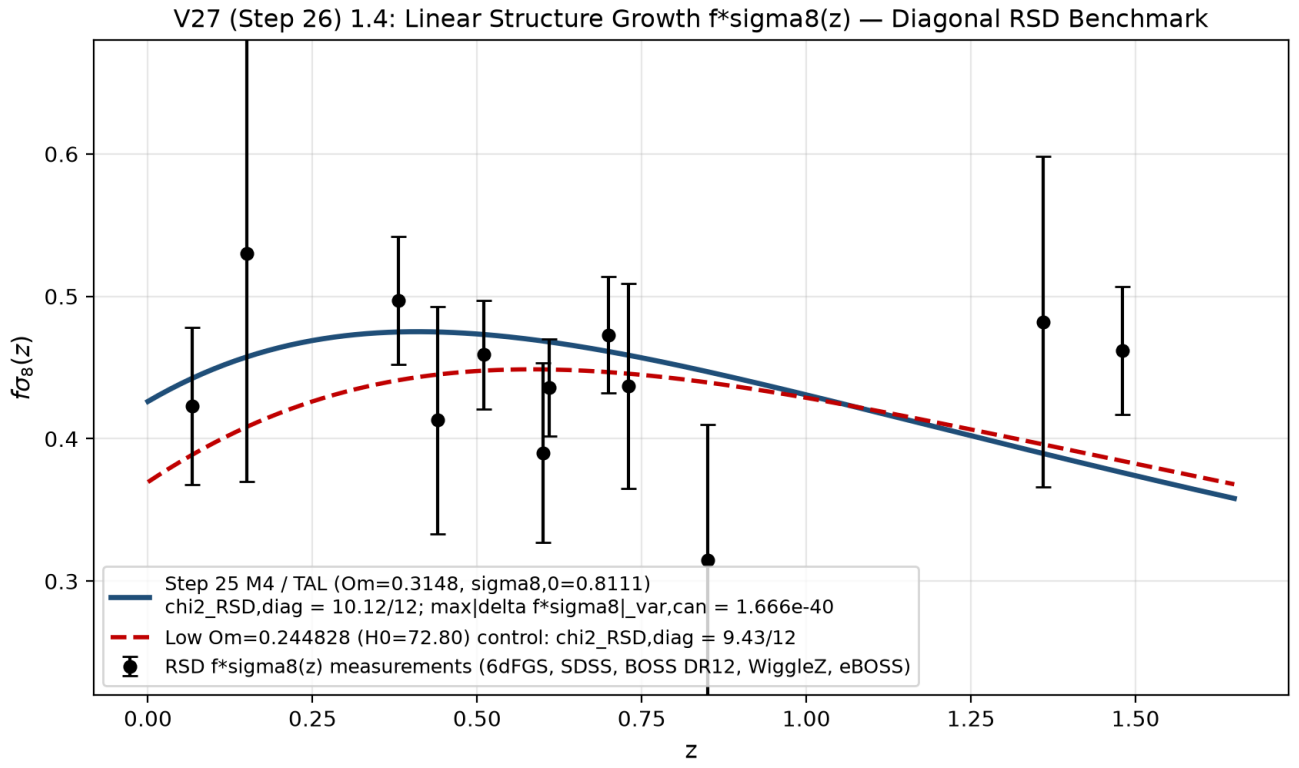


Figure 3: RSD $f\sigma_8(z)$ diagonal benchmark (`step1_4_rsd_fsigma8_plot_en.png`)

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source contains only dust matter, not the thermal-plasma trace. The pulls divide only by the quoted observational errors.

Proxies and sources (re-audit R6). The clock proxy $|\delta G/G| \approx 2|\phi|$ and the coefficients 0.08 ($\delta Y_p \approx 0.08\,\delta H/H$) and 1.6 ($\delta \ln(\mathrm{D/H}) \approx 1.6\,\delta H/H$) are **assumed order-of-magnitude proxies**, not derived here. Frame and sign (review F01): with $g_J = A^2 g_E$ the Jordan-frame Ricci term carries A^{-2}/G_* (A^{-2} multiplies the Planck mass squared), so the tensor coupling is $1/G_T = A^{-2}/G_*$, i.e. $G_T = G_* A^2 = G_* e^{2\alpha_Y \phi}$, and $\phi < 0$ **decreases** G_T. The tensor coupling G_T, the operationally measured $G_{\rm lab}$ (Section 1.5) and the bare $2|\phi|$ proxy are three different quantities; only the magnitude $2\alpha_Y|\phi|$ is used, and the sign and reference state are not propagated. For comparison, the Kneller–Steigman fits (New J. Phys. 6, 117, 2004) give $\delta Y_p \approx 0.16\,(S-1)$ and $\delta \ln(\mathrm{D/H}) \approx 1.6\,(S-1)$ with $S = H'/H$: the coefficient 1.6 matches, 0.08 is a factor 2 smaller (immaterial at $|\delta H/H| \lesssim 10^{-25}$). Input values: $Y_p^{\rm std} = 0.24709$ is the PRIMAT central value (Pitrou, Coc, Uzan & Vangioni 2018, Phys. Rep. 754, 1); $10^5\,\mathrm{D/H}^{\rm std} = 2.5147$ is an **arithmetic proxy** (review P02): a linearized benchmark value without verified single-paper provenance, not a sourced BBN prediction (PRIMAT 2018 gives 2.459); $Y_p^{\rm obs} = 0.245 \pm 0.003$ is the PDG 2024 BBN review recommendation (Fields, Molaro & Sarkar), consistent with Aver et al. 2021 (JCAP 03, 027: 0.2453 ± 0.0034); $10^5\,\mathrm{D/H}^{\rm obs} = 2.527 \pm 0.030$ is Cooke, Pettini & Steidel 2018 (ApJ 855, 102) — the CSV field name `DH_e5_obs_PDG` is a legacy label (PDG 2024 recommends 2.547 ± 0.025).

Epoch	z	$H/(c/L)$	ρ_m [kg/m^3]	$\lvert\bar\phi_{\rm attr}\rvert$	$\delta H/H$ (scalar energy)	$\delta G/G$ (clock)	$\lvert\delta Y_p\rvert$
end of BBN, $T \approx 0.02$ MeV	1.0×10^8	6.992×10^{-9}	2.686×10^{-3}	1.995×10^{-22}	1.705×10^{-27}	3.990×10^{-22}	1.596×10^{-23}
D bottleneck, $T \approx 0.09$ MeV	4.0×10^8	1.119×10^{-7}	1.719×10^{-1}	1.277×10^{-20}	2.728×10^{-26}	2.553×10^{-20}	1.021×10^{-21}
$z = 10^9$, $T \approx 0.23$ MeV	1.0×10^9	6.992×10^{-7}	2.686×10^{0}	1.995×10^{-19}	1.705×10^{-25}	3.990×10^{-19}	1.596×10^{-20}

- $Y_p = 0.24709$ versus 0.245 ± 0.003 (pull 0.70); $10^5\,\mathrm{D/H} = 2.5147$ versus 2.527 ± 0.030 (pull -0.41).
- Maximum values over the three epochs: $H/(c/L) \le 6.992 \times 10^{-7}$, $|\bar\phi_{\rm attr}| \le 1.995 \times 10^{-19}$, $\delta H/H \le 1.705 \times 10^{-25}$, $|\delta Y_p| \le 1.596 \times 10^{-20}$. The qualitative expectation of a tiny adiabatic correction survives; this is not a general BBN bound.

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7. Local Yukawa constant-density bulk benchmarks (`expm1`)

The rows are bulk products $|\phi|_{\rm bulk} = \alpha_Y \gamma_L \rho$ for constant density. The half-space edge value $|\phi|_{\rm bulk}/2$ for $R \gg L$ is given only where a boundary geometry is defined (mean solar density, mean Earth density, white dwarf interior; status `half-space boundary`). The homogeneous FLRW, cosmic void and Milky Way disk rows have no boundary geometry here: their edge cell is n.a. (CSV/JSON value null, `boundary_status` = `n.a. (no boundary geometry)`, edge ΔH_0 null); their bulk value is neither copied into the edge column nor halved (review N01). The values are not solved surface fields of the Sun or the Earth (that needs the real source and density profile). Clock factors use `expm1` to avoid round-off zeros; the apparent shift is $\Delta H_0^{\rm app} = 67.40\,\mathrm{expm1}(\alpha_Y|\phi|)$. The row "V21 top-hat interior external benchmark" ($|\phi| = 0.0412$) is an **external input, not reproduced here**: an interior value of the V21 top-hat fixed-point profile (SLy `gradient`, $r = 2.0046$ km, $x = 0.041239$; APR4 `mass`, $r = 2.7542$ km, $x = 0.041233$; Section 1.7), not a surface value (edge column n.a.) and not a solution of the present action.

Benchmark	ρ [kg/m^3]	$\lvert\phi\rvert_{\rm bulk} = \alpha_Y\gamma_L\rho$	$\lvert\phi\rvert$ ($R \gg L$ edge)	expm1($-\alpha_Y\lvert\phi\rvert$)	$\Delta H_0^{\rm app}$ [km/s/Mpc]	fraction of 5.4 gap
Homogeneous FLRW, $z = 0$	2.686×10^{-27}	1.995×10^{-46}	n.a.	-1.995×10^{-46}	1.345×10^{-44}	2.490×10^{-45}
Cosmic void ($0.2\rho_{m,0}$)	5.373×10^{-28}	3.990×10^{-47}	n.a.	-3.990×10^{-47}	2.689×10^{-45}	4.980×10^{-46}
Milky Way disk (average)	1.000×10^{-20}	7.426×10^{-40}	n.a.	-7.426×10^{-40}	5.005×10^{-38}	9.269×10^{-39}
Mean solar density (bulk)	1408.0000	1.046×10^{-16}	5.228×10^{-17}	-1.046×10^{-16}	7.047×10^{-15}	1.305×10^{-15}
Mean Earth density (bulk)	5514.0000	4.095×10^{-16}	2.047×10^{-16}	-4.095×10^{-16}	2.760×10^{-14}	5.111×10^{-15}
White dwarf interior (bulk)	1.000×10^9	7.426×10^{-11}	3.713×10^{-11}	-7.426×10^{-11}	5.005×10^{-9}	9.269×10^{-10}
V21 top-hat interior external benchmark	n.a.	0.0412	n.a.	-0.0404	2.8349	0.5250

- Mean solar density bulk: $\Delta H_0^{\rm app} = 7.047 \times 10^{-15}$ km/s/Mpc (not a mathematical zero).
- Only the external V21 interior benchmark reaches $\Delta H_0^{\rm app} = 2.835$ km/s/Mpc (fraction 0.525 of the 5.4 km/s/Mpc gap); this is a conditional illustration; identifying a local clock factor with the cosmological H_0 needs a separate derivation with a defined observer convention.

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8. No-go appendix

8.1 M2 / M3 branches

- `M2` (flat, uncompensated negative vacuum density): $E^2(0) = -0.37020$, so the declared normalization to a positive H_0 today is unphysical. The root of $E^2(z) = 0$ is $z \approx 0.295737$ in the full background (matter-only approximation: 0.295878). A negative vacuum energy is not excluded in general; this specific normalized point is.
- `M3` (open, $\Omega_K = 1.3702$), with its **own** sound horizons re-integrated with the M3 $H(z)$ (review F03): $r_s(z_*) = 144.36622618$ Mpc, $r_d(z_{\rm drag}) = 147.01159289$ Mpc, $D_M(z_*) = 34916.4$ Mpc, $100\theta_* = 0.41346227$ (formal geometric pull -2092.09 on the 1D θ_* prior, not a likelihood significance) and BAO $\chi^2 = 2201.29927$: a very poor acoustic-geometry benchmark. (The superseded values with the frozen M4 horizons were 0.41368156 and 2209.46156.)

8.2 The $67.36\,(1 + 1/4\pi)$ coincidence

$67.36\,(1 + 1/4\pi) = 72.7203$ km/s/Mpc and $67.36\,e^{1/4\pi} = 72.9394$ km/s/Mpc are correct arithmetic identities, but no dynamical H_0 correction follows from them: the clock factor is set by the solved potential $|\phi|$ (Sections 1 and 7), not by the bare coupling $\beta_{\rm Yuk}$.

8.3 Inverse distance ladder with profiled statistics

Scenario	H_0	Ω_m	Ω_K	r_d [Mpc]	$P = H_0 r_d$ [km/s]	$\chi^2_{\rm BAO}$ only (12×12)	Δr_d vs Planck [Mpc]
1. Fixed Step 25 M4 baseline	67.40	0.3148	0.0000	147.09	9914.02	20.785	+0.00
2. M4 shape, P profiled	72.80	0.3148	0.0000	137.77 ± 0.65	10029.65 ± 47.43	14.701	-9.32
3. Ω_m and P profiled	72.80	0.2939	0.0000	140.02 ± 1.74	10193.67 ± 126.47	12.741	-7.07
4. Joint BAO + Planck r_d + assumed local H_0 prior	70.81	0.2719	0.0000	147.03 (prior σ 0.26)	10410.24	15.674	-0.06
5. Rigid all-z $H_0 = 72.8$, unchanged r_d	72.80	0.3148	0.0000	147.09	10708.32	194.342	+0.00
6. Fixed-P illustration	72.80	0.3148	0.0000	136.18	9914.02	20.785	-10.91
7. Flat 72.8, $\Omega_m = 0.2448$, re-int. r_d	72.80	0.2448	0.0000	150.65	10967.41	70.304	+3.56
8. Open 72.8, fixed ω_{cb}	72.80	0.2698	0.0098	147.09	10708.27	61.206	+0.00

- The $\chi^2_{\rm BAO}$ column contains only the DESI 12 × 12 term. Row 4 adds the priors: $\chi^2_{\rm joint} = \chi^2_{\rm BAO} + \chi^2_{r_d\,\rm prior} + \chi^2_{H_0\,\rm prior} = 15.67431585 + 0.06234684 + 3.97660606 = 19.71326876$; its r_d entry quotes the **input prior error** ±0.26 Mpc, not a fitted uncertainty. Row 5 is the rigid all-z rescaling with unchanged r_d, not a late-transition model (Section 3).
- The fixed-P illustration (row 6: $P = 9914.02$ km/s, $r_d = 136.18$ Mpc) is a conditional arithmetic example; $P = 9914.02$ is not the BAO best-fit product, and its earlier sigma label is withdrawn.
- The profiled BAO requirement at $H_0 = 72.8$ is $r_d = 140.02 \pm 1.74$ Mpc (two-parameter control), and the joint BAO + Planck r_d + assumed local H_0 benchmark prior control gives $\Delta\chi^2 = 6.973$.
- The rigid rescaling at unchanged shape and r_d is strongly disfavoured by BAO ($\chi^2 = 194.342$, $p = 4.79 \times 10^{-35}$).

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9. What remains and what is not established

Remains:

- A rigid 8% increase of $H(z)$ at all redshifts with fixed $E(z)$ shape and unchanged r_d fits BAO very poorly.
- The bare $\beta_{\rm Yuk}$ is not equal to a solved scalar potential or a clock calibration shift.
- In the short-range adiabatic branch with negligible oscillation energy the late-time cosmological scalar correction is extremely small.
- The specific M2 normalization is unphysical and the M3 curvature is a poor acoustic-geometry benchmark.
- The profiled constant SNe zero-point offset and the H_0–M_B degeneracy are correct geometric illustrations.

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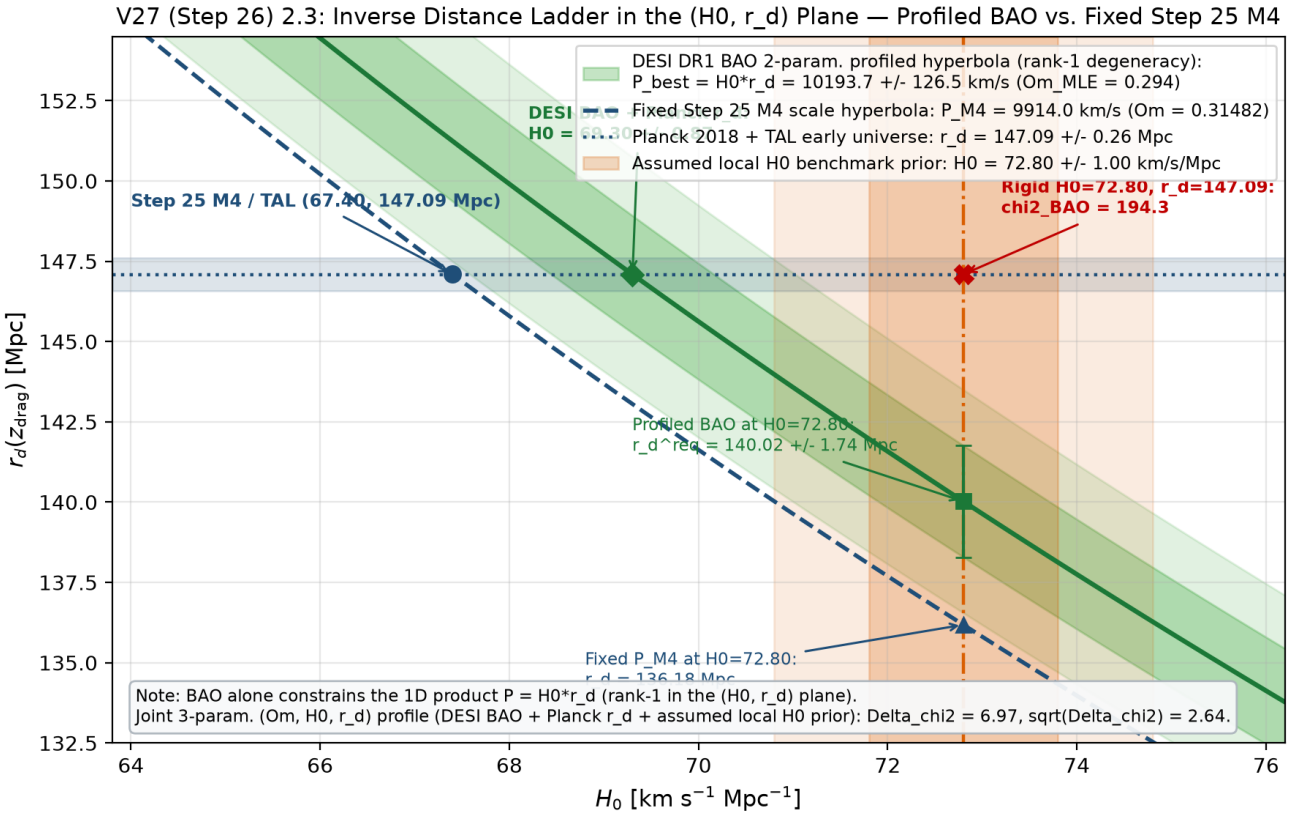


Figure 4: Inverse distance ladder in the (H_0, r_d) plane (`step2_3_inverse_ladder_rd_H0_exclusion_en.png`)

Not established in this package:

- a joint Planck + Pantheon+/Union3 + RSD + BBN data validation;
- general physical exclusion strengths of the kind formerly suggested by sigma labels;
- a cosmological closure of the scalar–tensor theory for arbitrary allowed initial conditions;
- a new TAL-specific empirical advantage over a properly fitted ΛCDM.

Closing statement. In the examined M4 branch with $L = 10$ km, fixed coupling and negligible free scalar oscillation energy, the local Yukawa factor provides no derived mechanism for a global Hubble constant of 72.8 km/s/Mpc. A simple global rescaling of a fixed early background and a fixed late-time shape is strongly disfavoured by BAO. The result is not a comprehensive joint Planck–SNe–RSD–BBN likelihood analysis and not a no-go theorem for all cosmological branches of the theory.

9.1 Nyitott korlátok / Open limitations

Item	Open limitation	Status in this package
P01	Verified primary source of the $z = 0.845$ ELG1 point 18.33 ± 0.62	chosen illustration without a verified primary source; not a DESI measurement; not in the 12-point likelihood
P02	Single-paper source of $10^5\,(\mathrm{D/H})_{\mathrm{std}} = 2.5147$	arithmetic proxy; an empirical standard-model test needs a verified input and theory error
P03	Coupled scalar–Friedmann–matter initial-value problem and free homogeneous amplitudes	branch $C = 0$ is selected explicitly; no statement about all initial conditions or all cosmological TAL branches
P04	Planck TT/TE/EE + lensing likelihood and a recombination calculation	3D prior assumed, z_* and z_{drag} fixed control parameters; geometric benchmark, not a CAMB/CLASS validation
P05	Real Pantheon+/Union3 data vector and covariance	synthetic 18-node metric, not an observational model preference
P06	RSD covariance, overlaps, AP mapping, derived amplitude, k spectrum	$k = 0.1\,h$/Mpc variation with fixed $\sigma_{8,0}$; 1.666×10^{-40} refers to this ODE, normalization and initial conditions
P07	BBN thermal trace, variable g_*, $e^{\pm}$ thermodynamics, reaction network	three sample epochs with assumed sensitivities; not an abundance likelihood
P08	Real solar/terrestrial density profiles, surface Yukawa solution, link of local clocks to the distance ladder	bulk and half-space limits; the $\Delta H_0^{\mathrm{app}}$ table is an arithmetic illustration

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Item	Open limitation	Status in this package
P09	The V26 book as a whole, all historical conventions and all stellar BVPs	this review concerns Step 26; the V26 normalization identity is a targeted check only

10. Files and reproducibility

- `desi_2024_gaussian_bao_ALL_GCcomb_mean.txt`: 376 B, SHA-256 `dd2873a0b88459a491af3c0c0307ba059f62df9211d5b976760f310565a1be68`
- `desi_2024_gaussian_bao_ALL_GCcomb_cov.txt`: 2,170 B, SHA-256 `bbafa9074b51cf1a45e0d10e4f37db8c0e80a5d1d1788857abb7fc49fb21abcc`
- `tal_cosmology_common.py`: 24,847 B, SHA-256 `54c5636b35631af314f501060e39adf723a46cada78db15526b0591a87c2c109`
- `tal_hubble_desi_bao_summary.json`: 52,790 B, SHA-256 `94ba04e4c09fc8a7d6a94eed04eecf03c9f794444a3a9f646a58f80aa1e3ca53`
- Generators: `tal_cosmology_common.py` (shared background), `run_step1_1.py`, `run_step1_2_and_1_3.py`, `run_step1_4_to_1_6.py`, `run_step2_nogo_and_report.py`, `run_step26_revision_controls.py` (re-audit controls), `hubble_desi_bao_tal_audit.py`; text: `generate_step26_markdown.py` + `step26_md_templates.py`, text regression test `test_step26_text_regression.py`; PDF: `compile_step26_clean_pdfs.py`; runner: `run_all.py` (order and data hash check, `--skip-block1`), dependencies: `requirements.txt`.
- The package is self-contained: both DESI input files, all CSV and PNG outputs, the strict JSON (`allow_nan=False`, NaN written as `null`) and `SHA256SUMS.txt` are included.
- External sources: DESI VI (arXiv:2404.03002), DESI III (arXiv:2404.03000), Planck 2018 VI (arXiv:1807.06209), Cobaya `bao_data` repository (DESI 2024 Gaussian BAO mean and covariance).

11. Revision log

Item	Change	Where
R1	Kinetic normalization reconciled with V26 (file:line), $\phi_{\rm old} = 2\phi_{\rm new}$ needs $\alpha_{\rm new} = 2\alpha_{\rm old}$; carried-over results listed; compact-star $\lvert\phi\rvert$ marked as external V21 interior value; $2\alpha^2\beta$ relabelled sensitivity control	1.5, 1.7, 7
R2	Piecewise $H = sH_{\rm M4}$ ($z < z_t$) control for $z_t \in \{2.5, 10, 100, 1000\}$; D_M/s row reclassified as inconsistent geometric illustration	3, `step26_r2_piecewise_transition.csv`
R3	Self-contained package: DESI inputs with hashes, all CSV/PNG, `requirements.txt`, `run_all.py`, strict JSON; old `build_step26_pdfs.py` moved out of the package	10, README
R4	Joint χ^2 split into BAO, r_d-prior and H_0-prior terms; table column is BAO-only; r_d ± 0.26 labelled input prior error	2, 8.3
R5	Symmetric zero-point offset for M4 and the low-Ω_m model; lowest-node value separated from the $5\log_{10}$ identity	4, `step26_r5_sne_symmetric_offsets.csv`
R6	BBN clock proxy and coefficients labelled assumed proxies; frame and sign discussed; sources given	6
5.1	Float64 branch uses $1 + \Delta G_{\rm eff,can}/G_N$; absolute-value headers; independent rescaled check quoted	5
5.2	$z = 0$ drift separated from the Fermi–Dirac interpolation error, which is computed	1.6
5.3	θ_*-matched model relabelled to $\Omega_m = 0.24419$, $\omega_{cb} = 0.12877$; inverse-ladder figure uses $\sqrt{\Delta\chi^2}$ wording; provenance statement reworded	3, 8.3, header

12. Revíziós napló 2 / Revision log 2 (second review, F01–F12)

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Item	Change	Targeted check
F01	$G_J = GA^{-2}$ replaced by $1/G_T = A^{-2}/G_*$, $G_T = G_* A^2$; $\phi < 0$ decreases G_T; G_T, $G_{\rm lab}$ and the bare $2\lvert\phi\rvert$ proxy separated (Section 6, code comments, R6 metadata)	text and JSON `R6_bbn_proxy_provenance.clock_proxy` contain the new relation
F02	G dictionary after Section 1.5 (G_*, G_T, $G_{\rm lab} \approx G_*(1+q)$, $G_N := G_*$)	$1/(1+q) = 0.9262883$ computed in `run_step26_revision_controls.py`
F03	M3 uses its own r_s, r_d	$r_s = 144.36622618$, $r_d = 147.01159289$ Mpc, $100\theta_* = 0.41346227$, $\chi^2_{\rm BAO} = 2201.29927$ (assertion against the review values)
F04	step2 status strings and CLI text rewritten: "BAO fit", "deviation on illustrative prior", "pull vs fixed target"; no Planck exclusion or curvature-likelihood wording	automatic string scan in `run_step26_revision_controls.py`
F05	72.8 ± 1.0 labelled assumed local H_0 benchmark prior; JSON key `H0_local_benchmark_prior_km_s_Mpc`; figure labels changed	labels no longer attribute the value to SH0ES (CSV/JSON/PNG text)
F06	Section 1.5 sentence on the $2\alpha_Y^2\beta_{\rm Yuk}$ row	text check
F07	RSD legend shows the canonical maximum	1.6664×10^{-40} in the legend and JSON
F08	lowest-node sentence removed from Section 4, identity kept	text check
F09	SNe figure: low-Ω_m offset curve added, "profiled constant zero-point offset", $N = 18$, $\nu = 17$	low-Ω_m with offset $\chi^2 = 22.777527$
F10	row renamed "V21 top-hat interior external benchmark", edge column n.a., provenance added	SLy $r = 2.0046409680935633$ km, $x = 0.041239199439254325$; APR4 $r = 2.75417748649486$ km, $x = 0.041233030079658145$
F11	PDF builder and `run_all.py --pdf` exit nonzero without pandoc/pdflatex or without a newly written PDF; `--tex-only` has its own status	test with a `PATH` lacking pdflatex
F12	`theta_star_100_rs_only_shifted_fixed_DM` (renamed) and `theta_star_100_both_endpoints_shifted` (new)	low-Ω_m: 1.04248582 and 1.04250287

13. Revíziós napló 3 / Revision log 3 (third re-audit, N01, N02, T01)

Item	Change	Targeted check
N01	Section 7 table: the "$R \gg L$ edge" cell of the homogeneous FLRW, cosmic void and Milky Way disk rows is n.a. (no boundary geometry); `step1_6_local_yukawa_environments.csv` and the JSON carry null in `abs_phi_surface_R_gg_L` and `Delta_H0_apparent_surf_km_s_Mpc` and the new per-row field `boundary_status`; bulk values unchanged (not halved)	assertion in `run_step26_revision_controls.py` (JSON `review3_N01_N02_T01`); text regression test
N02	Section 1.7 "Carried over unchanged": the parenthesis now separates the background scalar-energy correction $\Omega_\phi \sim 10^{-47}$ from the direct Yukawa perturbation of RSD ($\delta G_{\rm eff}/G_* \approx 3.8 \times 10^{-41}$, $\delta f\sigma_8 \approx 1.67 \times 10^{-40}$, Section 5); numbers taken from the JSON	text regression test (HU and EN sentence)
T01	New `test_step26_text_regression.py`, run by `run_all.py` after the Markdown generation, nonzero exit on failure: checks the HU/EN F06 sentence, the F08 identity wording, the absence of the superseded wrong sentences (V26 factor-2 growth-law attribution; the identity value attributed to the lowest table node), the N01 table cells, the N02 sentence and the canonical legend string of the RSD figure source (F07); the comment in `run_step26_revision_controls.py` updated	mutation test on a scratch copy: reinserting a wrong sentence gives a nonzero exit

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